hw10

Problem 1. Divisibility

1. first we divide 1001 by 23:

$$23 \cdot 43 = 989, \qquad 1001 - 989 = 12$$

   so

$$1001 = 23 \cdot 43 + 12,$$

   so the quotient is $\boxed{43}$ and the remainder is $\boxed{12}$.

2. next we divide -512 by 15

   We want integers q, r where

$$-512 = 15q + r, \qquad 0 \leq r < 15.$$

   so

$$15(-35) = -525, \qquad -512 - (-525) = 13,$$

   we get

$$-512 = 15(-35) + 13.$$

   so the quotient is $\boxed{-35}$ and the remainder is $\boxed{13}$

2.

1. For 360:

$$360 = 36 \cdot 10 = (2^2 \cdot 3^2)(2 \cdot 5) = 2^3 \cdot 3^2 \cdot 5$$

   so

$$\boxed{360 = 2^3 \cdot 3^2 \cdot 5.}$$

2. and since for 2310:

$$2310 = 231 \cdot 10 = (3 \cdot 7 \cdot 11)(2 \cdot 5)$$

   so

$$\boxed{2310 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11.}$$

3. and for 11!:

   we count the exponent of each prime $p \leq 11$

   and for 2:
$$\left\lfloor \frac{11}{2} \right\rfloor + \left\lfloor \frac{11}{4} \right\rfloor + \left\lfloor \frac{11}{8} \right\rfloor = 5 + 2 + 1 = 8.$$

   for 3:
$$\left\lfloor \frac{11}{3} \right\rfloor + \left\lfloor \frac{11}{9} \right\rfloor = 3 + 1 = 4.$$

   For 5:
$$\left\lfloor \frac{11}{5} \right\rfloor = 2.$$

   for 7:
$$\left\lfloor \frac{11}{7} \right\rfloor = 1.$$

   For 11:
$$\left\lfloor \frac{11}{11} \right\rfloor = 1.$$

   So
$$\boxed{11! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7 \cdot 11.}$$

3.

Given
$$a = 2^3 \cdot 3^4 \cdot 5^2 \cdot 11^3 \qquad \text{and} \qquad b = 2^5 \cdot 3^2 \cdot 7^2 \cdot 11.$$

Using the prime-factor formulas:

$$\gcd(a, b) = 2^{\min(3,5)} \cdot 3^{\min(4,2)} \cdot 5^{\min(2,0)} \cdot 7^{\min(0,2)} \cdot 11^{\min(3,1)}.$$

Thus
$$\gcd(a, b) = 2^3 \cdot 3^2 \cdot 11 = 8 \cdot 9 \cdot 11 = \boxed{792}.$$

Similarly,
$$\operatorname{lcm}(a, b) = 2^{\max(3,5)} \cdot 3^{\max(4,2)} \cdot 5^{\max(2,0)} \cdot 7^{\max(0,2)} \cdot 11^{\max(3,1)}.$$

So
$$\operatorname{lcm}(a, b) = 2^5 \cdot 3^4 \cdot 5^2 \cdot 7^2 \cdot 11^3 = \boxed{4226191200}.$$

Now verify $\gcd(a, b) \operatorname{lcm}(a, b) = ab$:
$$\gcd(a, b) \operatorname{lcm}(a, b) = \left(2^3 \cdot 3^2 \cdot 11\right) \left(2^5 \cdot 3^4 \cdot 5^2 \cdot 7^2 \cdot 11^3\right) = 2^8 \cdot 3^6 \cdot 5^2 \cdot 7^2 \cdot 11^4.$$

Also,
$$ab = \left(2^3 \cdot 3^4 \cdot 5^2 \cdot 11^3\right) \left(2^5 \cdot 3^2 \cdot 7^2 \cdot 11\right) = 2^8 \cdot 3^6 \cdot 5^2 \cdot 7^2 \cdot 11^4.$$

Hence
$$\boxed{\gcd(a, b) \operatorname{lcm}(a, b) = ab.}$$

Problem 4. Extended Euclid Algorithm

We ask whether 103 has a multiplicative inverse in $\mathbb{Z}_{128}$. Such an inverse exists iff $\gcd(103, 128) = 1$.

Use Euclid's algorithm:

$$\begin{aligned} 128 &= 1 \cdot 103 + 25, \\ 103 &= 4 \cdot 25 + 3, \\ 25 &= 8 \cdot 3 + 1, \\ 3 &= 3 \cdot 1 + 0. \end{aligned}$$

So

$$\gcd(103, 128) = 1,$$

therefore the inverse exists.

Now back-substitute:

$$1 = 25 - 8 \cdot 3.$$

Since $3 = 103 - 4 \cdot 25$, we get

$$1 = 25 - 8(103 - 4 \cdot 25) = 33 \cdot 25 - 8 \cdot 103.$$

Since $25 = 128 - 103$, we get

$$1 = 33(128 - 103) - 8 \cdot 103 = 33 \cdot 128 - 41 \cdot 103.$$

Thus

$$-41 \cdot 103 \equiv 1 \pmod{128}.$$

Therefore the inverse of 103 modulo 128 is

$$\boxed{-41 \equiv 87 \pmod{128}.}$$

Problem 5. Euclid's Algorithm and Bézout's Theorem

(a) Compute $\gcd(540, 168)$.

$$\begin{aligned} 540 &= 3 \cdot 168 + 36, \\ 168 &= 4 \cdot 36 + 24, \\ 36 &= 1 \cdot 24 + 12, \\ 24 &= 2 \cdot 12 + 0. \end{aligned}$$

So

$$\boxed{\gcd(540, 168) = 12.}$$

(b) Find integers x, y such that

$$540x + 168y = 12.$$

Back-substitute:

$$12 = 36 - 24.$$

Since $24 = 168 - 4 \cdot 36$,

$$12 = 36 - (168 - 4 \cdot 36) = 5 \cdot 36 - 168.$$

Since $36 = 540 - 3 \cdot 168$,

$$12 = 5(540 - 3 \cdot 168) - 168 = 5 \cdot 540 - 16 \cdot 168.$$

Thus one solution is

$$\boxed{x = 5, \qquad y = -16.}$$

Problem 6. Co-primality

We must show that for every integer $k \geq 1$, the numbers $3k + 2$ and $4k + 3$ are relatively prime.

Consider the linear combination

$$4(3k + 2) - 3(4k + 3) = 12k + 8 - 12k - 9 = -1.$$

If a positive integer d divides both $3k + 2$ and $4k + 3$, then d must divide every integer linear combination of them. In particular, $d \mid (-1)$, so $d = 1$.

Therefore

$$\boxed{\gcd(3k + 2,\, 4k + 3) = 1}$$

for every integer $k \geq 1$. Hence $3k + 2$ and $4k + 3$ are relatively prime.